

Artifact-Conditioned Interval Diagnostics for Flow-Matching Neural Posterior Estimation in a Controlled Gravitational-Wave Benchmark

Zhi Luo^{1,*} and Qi-Qin Jing²

¹*Department of Physics, Chongqing University, Chongqing 401331, China*

²*School of Physics and Astronomy, China West Normal University, Nanchong 637002, China*

(Dated: June 16, 2026)

Calibration checks for neural posterior estimators in gravitational-wave inference should remain interpretable when observations contain data-quality artifacts. We study marginal interval calibration in a controlled frequency-domain binary-black-hole benchmark with synthetic glitches, frequency masks, and power spectral density (PSD) mismatch. The posterior sampler is a support-aware flow-matching posterior estimator (FMPE) with a circular representation of coalescence phase. We compare raw marginal credible intervals with global rescaling, oracle artifact-stratified rescaling, hard predicted-label rescaling, and soft learned artifact-aware interval rescaling (LAIR). In the 1024-bin evaluation, a single global scale fitted on mixed calibration data transfers poorly to frequency-mask cases, giving a mean absolute 90% marginal coverage error (MA90CE) of 0.1195. Soft LAIR lowers the corresponding error to 0.0672, although it is not uniformly better than the raw FMPE intervals. A 40-split LAIR evaluation and a six-checkpoint FMPE training-seed study show that the frequency-mask behavior is not a single-split artifact. The classifier recognizes frequency masks and PSD mismatch reliably, while glitch recall remains low. Waveform-resolution tests, PyCBC/LAL TaylorF2 backend checks, prior and Gaussian baselines, and controlled-likelihood reference-posterior probes indicate that marginal coverage must be read together with posterior width, geometry, and likelihood-based diagnostics. These results support using LAIR as an artifact-structured interval diagnostic, not as a substitute for posterior validation.

I. INTRODUCTION

Gravitational-wave parameter estimation infers compact-binary source parameters from noisy detector strain. Likelihood-based analyses provide the standard reference for precision inference, but repeated waveform evaluations and stochastic sampling can be costly. Neural posterior estimation reduces this cost by amortizing the map from strain data to posterior samples.

Good calibration on clean simulations is not sufficient when the data distribution changes. Detector data can include nonstationary noise, missing frequency bands, glitches, power spectral density (PSD) drift, calibration uncertainty, and other effects absent from an idealized training set. Such effects can produce artifact-specific coverage errors. At the same time, acceptable marginal coverage is not by itself evidence of an informative posterior, since broad intervals can cover the truth while carrying little useful structure.

We use a controlled frequency-domain binary-black-hole benchmark to isolate these effects. The simulator has a TaylorF2-like nonspinning waveform, analytic whitening, two detector channels, and known synthetic artifact labels. The posterior estimator is a conditional flow-matching posterior estimator (FMPE). Bounded parameters are trained in stable logit coordinates, and coalescence phase is represented by sine and cosine coordinates to avoid the artificial $0-2\pi$ discontinuity.

The diagnostic considered here is learned artifact-aware interval rescaling (LAIR). Calibration events are

used to fit marginal interval scale factors for each artifact family and parameter. At evaluation time an artifact classifier returns probabilities $p_\psi(a | x)$, and the deployed scale for parameter k is the probability-weighted average of the artifact-specific scales. The procedure uses predicted artifact information for evaluation events; it does not require simulator labels for those events.

The analysis asks whether artifact-conditioned interval scaling exposes failure modes that are hidden by a single global calibration. We evaluate the main 1024-bin FMPE checkpoint, repeat the LAIR calculation across 40 independent simulator/calibration splits, and assess six independently trained 1024-bin FMPE checkpoints. Classifier comparisons, waveform-resolution tests, simple posterior baselines, and controlled-likelihood reference probes are included so that the coverage results are not interpreted in isolation.

Figure 1 gives the workflow. FMPE produces posterior samples, after which the LAIR-family procedures rescale marginal intervals and report artifact-conditioned coverage.

A. Relation to Prior Work

Neural posterior estimation has become an important route to faster gravitational-wave inference. Autoregressive normalizing flows showed that neural density estimators can learn compact-binary posteriors in controlled settings [1]. DINGO-style systems extended this approach to rapid gravitational-wave analyses and added importance-sampling corrections for amortized proposals [2, 3]. Flow matching formulates generation through a

* Corresponding author: zhiluo@cqu.edu.cn

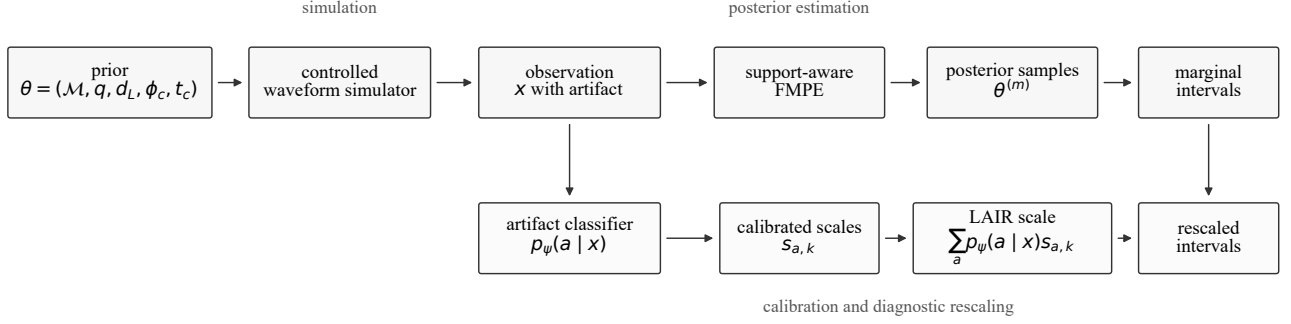


FIG. 1. Inference and diagnostic workflow. Simulated observations are passed to a support-aware flow-matching posterior estimator. Marginal intervals are then evaluated raw or after global, oracle artifact-stratified, hard predicted-label, or soft LAIR rescaling.

learned continuous vector field [4], and has been adapted to simulation-based inference [5]. The estimator used here belongs to this flow-matching family, with the artifact label available only through the controlled simulator.

Posterior validation is a separate issue. Simulation-based calibration tests rank uniformity under the assumed simulator [6], and TARP-style diagnostics provide related sampling-based checks [7]. Marginal credible-interval coverage is therefore used here as a focused diagnostic. It can reveal artifact-dependent miscalibration, but it does not determine the joint posterior, tails, or information content.

Likelihood-based parameter-estimation tools such as Bilby and PyCBC Inference remain the standard reference implementations for production analyses [8, 9]; LALSuite supplies core waveform and data-analysis infrastructure [10]. Recent simulation-based studies have examined robustness to real noise and missing information [11, 12], and public LIGO-Virgo-KAGRA (LVK) open data make such questions directly testable [13, 14]. The experiments below do not use LVK strain data. They instead use known artifact families so that mode-resolved calibration errors can be measured directly.

1. Position relative to DINGO, FMPE, and likelihood-corrected neural inference

DINGO is the closest established neural posterior-estimation benchmark family for gravitational-wave inference. Its importance-sampling correction also illustrates how an amortized proposal can be checked against a likelihood. The present work does not implement that correction and does not run DINGO, Bilby, or PyCBC inference. The comparison is therefore limited in scope: LAIR is tested as a marginal artifact-conditioned interval diagnostic within a shared controlled simulator. The conditional RealNVP result in Sec. III C is included only as a lightweight discrete-flow comparator under the same simulator and metrics.

2. Model-family choice: discrete flows, flow matching, and score-based diffusion

Discrete normalizing flows are the established neural-density baseline for gravitational-wave parameter estimation. DINGO-style conditional flows show that amortized density estimators can produce rapid compact-binary posterior samples. The RealNVP baseline below tests this discrete-flow family only within our simulator; it is not a reproduction of DINGO or DINGO-IS.

The main estimator uses flow matching. Rather than applying a finite sequence of invertible maps, the model learns a continuous vector field that transports a simple base distribution to the observation-conditioned posterior target. This formulation is compatible with simulation-based inference and with the support-aware and circular parameterization used in the benchmark.

Score-based diffusion is a natural alternative for conditional posterior sampling, but it is outside the present comparison. A fair diffusion study would require a separate sampler, runtime tuning, and calibration analysis under the same metrics.

II. CONTROLLED INFERENCE PROBLEM AND METHODS

A. Parameters and Priors

The physical parameter vector is

$$\theta = (\mathcal{M}, q, d_L, \phi_c, t_c),$$

where $\mathcal{M}$ is detector-frame chirp mass, $q = m_2/m_1 \leq 1$ is the mass ratio, d_L is luminosity distance, ϕ_c is coalescence phase, and t_c is coalescence time. The symmetric mass ratio and total mass are

$$\eta = \frac{q}{(1+q)^2}, \quad M = \mathcal{M}\eta^{-3/5}.$$

In this paper $\mathcal{M}$ denotes chirp mass and M denotes total mass. The priors are independent and uniform:

$$\mathcal{M} \in [8, 18] M_{\odot}, \quad q \in [0.55, 1], \quad d_L \in [300, 1800] \text{ Mpc},$$

$$\phi_c \in [0, 2\pi), \quad t_c \in [-0.035, 0.035] \text{ s}.$$

The prior volume is deliberately compact: it spans mass, distance, phase, and time uncertainty while keeping repeated single-GPU experiments feasible.

B. Signal and Observation Model

The simulator uses a transparent TaylorF2-like frequency-domain inspiral model rather than a production waveform approximant. The complex strain scales schematically as

$$\tilde{h}(f; \theta) \propto \frac{\mathcal{M}^{5/6}}{d_L} f^{-7/6} \exp\{i\Psi(f; \theta)\},$$

with phase contributions from coalescence time, coalescence phase, and a nonspinning post-Newtonian expansion. A smooth taper is applied near the inspiral cutoff. Each observation contains two detector channels represented by real and imaginary frequency-domain components after analytic whitening, together with a mask channel.

This setup is a controlled inverse problem, not a full waveform model. It omits spins, precession, higher modes, sky localization, inclination, calibration uncertainty, and real detector nonstationarity. The restriction keeps the prior-generating process and artifact label known exactly. The PyCBC/LAL TaylorF2 comparison below checks phase, amplitude, and the fixed detector-projection conventions used in this simulator.

C. Controlled Likelihood and Whitening

The reference-posterior probes use the Gaussian likelihood implied by the whitened simulator. For detector channel I and frequency bin j ,

$$d_{Ij} = m_j \left[\tilde{h}_I(f_j; \theta) + n_{Ij} \right],$$

where $m_j \in \{0, 1\}$ is the frequency mask and the whitened noise satisfies $n_{Ij} \sim \mathcal{N}(0, 1)$ independently for real and imaginary components. The controlled log likelihood is

$$\log p(d | \theta) = -\frac{1}{2} \sum_{I,j} m_j \left| d_{Ij} - \tilde{h}_I(f_j; \theta) \right|^2 + \text{const.}$$

PSD factors are absorbed into the whitening operation. The two detector channels use fixed gains and time delays rather than sky-dependent antenna patterns, so the

problem remains a controlled two-channel inverse problem rather than a production detector response model.

PSD mismatch is introduced by changing the whitening tilt used to generate an evaluation observation while retaining the nominal whitening convention for the trained estimator and controlled reference model. Frequency-mask artifacts set a contiguous band of m_j to zero and expose the missingness through the mask channel. Glitch artifacts add a localized sine-Gaussian-like disturbance after the clean signal is generated. These definitions make the artifact label and the likelihood mismatch known by construction.

D. Artifact Families

The label set is

$$a \in \{\text{clean}, \text{glitch}, \text{frequency mask}, \text{PSD mismatch}\}.$$

Clean observations use the nominal simulator. Glitch observations contain a localized sine-Gaussian-like disturbance. Frequency-mask observations remove a band of frequency bins and expose the missingness through the mask channel. PSD-mismatch observations perturb the whitening model.

In mixed evaluation, a clean observation is drawn with probability 0.65 and an artifact observation with probability 0.35. Conditional on an artifact, the three artifact families are sampled uniformly. A 4096-event sample gives the realized fractions in Table I and Fig. 2. Qualitative examples appear in Fig. 3.

TABLE I. Realized mixed-mode artifact composition from 4096 simulated events.

Artifact	Count	Fraction
clean	2632	0.6426
glitch	463	0.1130
frequency mask	522	0.1274
PSD mismatch	479	0.1169

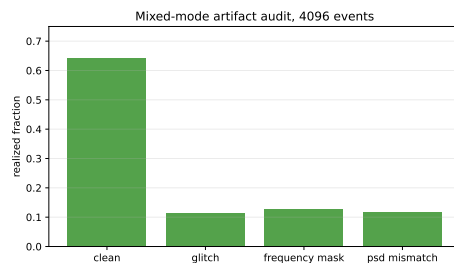


FIG. 2. Realized artifact counts in a 4096-event mixed-mode sample. The clean fraction follows the configured 0.65 probability, and the artifact families are approximately balanced within the remaining probability mass.

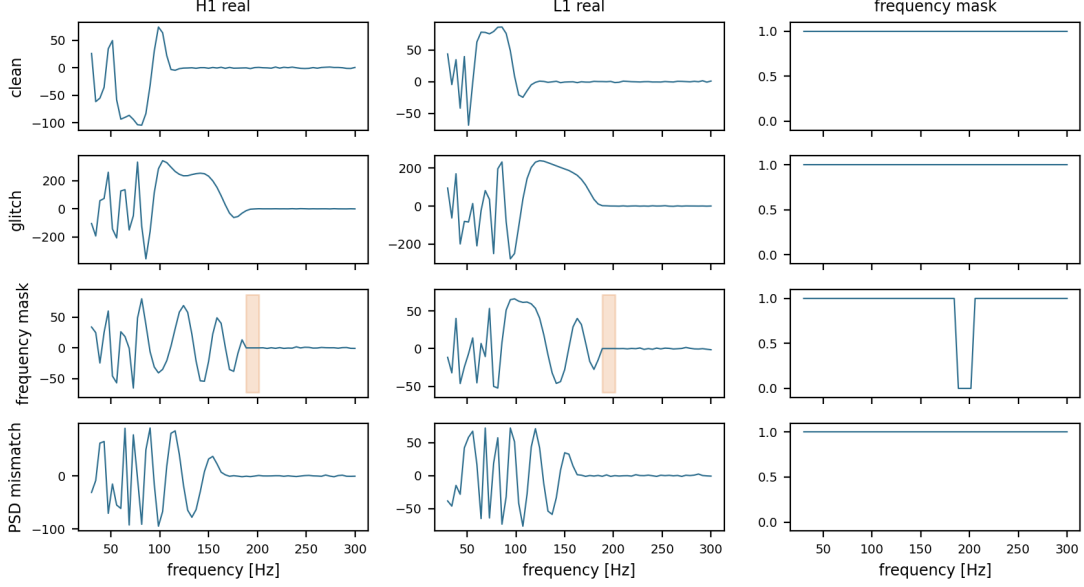


FIG. 3. Representative synthetic artifact examples. Frequency masks affect the mask channel directly; glitches and PSD mismatch appear as subtler frequency-dependent structure.

E. Posterior Estimator and Calibration Diagnostics

1. Support-Aware Transforms

For each bounded scalar parameter $x \in [\ell, u]$, the model uses

$$z = \frac{x - \ell}{u - \ell}, \quad y = \log \frac{z}{1 - z},$$

with numerical clipping before the logit. The inverse transform maps samples back into the physical prior interval. This keeps decoded $\mathcal{M}$, q , d_L , and t_c samples inside the prior support. Unless stated otherwise, interval rescaling for bounded scalar parameters is performed in this transformed coordinate: endpoints are mapped through the box logit, widened or narrowed about the transformed posterior center, and decoded back to the physical interval. This convention avoids unnecessary asymmetry near hard prior boundaries.

2. Circular Phase Treatment

Coalescence phase is embedded as

$$(\cos \phi_c, \sin \phi_c)$$

during training and decoded with atan2 modulo 2π . The model target is therefore

$$y(\theta) = (\text{logit } \mathcal{M}, \text{logit } q, \text{logit } d_L, \cos \phi_c, \sin \phi_c, \text{logit } t_c).$$

The phase coordinates are decoded after normalization by $r_\phi = \sqrt{u_\phi^2 + v_\phi^2}$, where (u_ϕ, v_ϕ) is the generated phase

embedding. The implementation records r_ϕ as an off-circle diagnostic. Large departures from unity indicate that the Euclidean flow has moved away from the training manifold even when the decoded angle is defined. Evaluation uses circular phase errors and geodesic intervals on S^1 . The wrapped phase residual is

$$\Delta\phi = \text{atan2}(\sin(\phi - \phi^*), \cos(\phi - \phi^*)).$$

For an interval with circular center $\bar{\phi}$ and half-width $h \leq \pi$, the truth is covered when

$$|\text{atan2}(\sin(\phi^* - \bar{\phi}), \cos(\phi^* - \bar{\phi}))| \leq h.$$

This definition removes the discontinuity at the $0-2\pi$ boundary.

3. Conditional Flow Matching

Let x denote the observation and y the support-aware target. A base sample z_0 is drawn from a standard Gaussian. Conditional flow matching trains a time-dependent vector field $v_\varphi(z, t, x)$ using the straight-line interpolation

$$z_t = (1 - t)z_0 + ty,$$

with target velocity $y - z_0$. The objective is

$$\mathcal{L}(\varphi) = \mathbb{E}_{x, y, z_0, t} \left[\|v_\varphi(z_t, t, x) - (y - z_0)\|_2^2 \right].$$

At inference time the learned ordinary differential equation is integrated from the Gaussian base distribution to $t = 1$, and generated samples are decoded to normalized physical coordinates for evaluation.

4. Marginal Interval Rescaling

Raw posterior samples define the marginal credible intervals. Global rescaling fits one scale factor per parameter on calibration data and applies it to all evaluation intervals. Oracle artifact-stratified rescaling fits one scale per artifact and parameter, then uses the true artifact label. Predicted-label rescaling replaces the true label with the classifier’s hard prediction. These operations affect marginal interval widths only; they do not move posterior centers or reconstruct joint correlations. For artifact class a , parameter k , and nominal levels $\alpha \in \mathcal{A}$, the calibrated scale is

$$s_{a,k} = \arg \min_{s>0} \sum_{\alpha \in \mathcal{A}} w_{\alpha} |\hat{c}_{\text{cal}}(a, k, \alpha; s) - \alpha|,$$

where $\hat{c}_{\text{cal}}$ is empirical calibration coverage after multiplying the marginal half-width by s . Global rescaling uses the same objective after pooling artifact classes. The primary scalar metric below is based on $\alpha = 0.9$, while coverage curves evaluate the fitted intervals over a grid of nominal levels.

5. Artifact Classifier and LAIR

The artifact classifier is a one-dimensional convolutional network over the frequency axis. Its channels are detector real and imaginary components plus the mask. Let $p_{\psi}(a | x)$ be the classifier probability for artifact class a . LAIR fits scale factors $s_{a,k}$ for artifact class a and parameter k on calibration data. The soft deployed scale is

$$s_k(x) = \sum_a p_{\psi}(a | x) s_{a,k}.$$

The rescaled marginal interval is centered on the event’s posterior center, with its half-width multiplied by $s_k(x)$.

This implementation of LAIR is an artifact-conditioned marginal interval diagnostic. It rescales intervals but leaves posterior centers and joint sample geometry unchanged, so its output must be interpreted together with the raw samples and reference diagnostics.

6. Baselines

Two baselines contextualize the coverage results. A prior-only baseline asks how often the truth falls in central prior intervals. A diagonal Gaussian posterior baseline gives a simple parametric approximation in the 256-bin low-resolution setting. Low MA90CE for either baseline does not imply an accurate posterior; it can also reflect broad intervals.

7. Metrics

For nominal level α , the empirical marginal coverage for parameter k is

$$\hat{c}_k(\alpha) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\theta_{i,k}^{\text{true}} \in I_{i,k}^{\alpha}\}.$$

The primary scalar metric is mean absolute 90% marginal coverage error,

$$\text{MA90CE} = \frac{1}{K} \sum_{k=1}^K |\hat{c}_k(0.9) - 0.9|.$$

We also track normalized mean absolute error, bias, posterior widths, and coverage curves. Event-bootstrap intervals resample evaluation events for a fixed trained model and fixed calibration procedure.

III. EXPERIMENTS AND RESULTS

All experiments use the controlled simulator described above. The FMPE tiers are summarized in Table II. The 1024-bin tier is the main high-resolution checkpoint. The 2048-bin tier tests frequency resolution at a smaller evaluation budget. The 256-bin tier is retained as a low-resolution check for earlier LAIR and baseline diagnostics.

The 1024-bin LAIR evaluation uses the trained 1024-bin checkpoint, a matching 1024-bin artifact classifier, 256 mixed calibration events, 256 evaluation events per mode, 512 posterior samples per event, 48 ODE steps, and 200 bootstrap resamples over evaluation events. A repeated evaluation reuses the same FMPE checkpoint across 40 independent simulator/calibration splits. That run measures evaluation-split and calibration-split variability; it is not a multi-seed FMPE training study. The classifier is trained on balanced simulated artifact batches and evaluated on 2048 balanced validation events. All figures and tables are generated from retained metric files.

A. Waveform-Resolution Diagnostic

The waveform-resolution diagnostic compares frequency grids from 64 to 2048 bins with a 4096-bin reference over 30–300 Hz using 128 prior draws. The mismatch is maximized over constant phase but not over time shifts.

Table III and Fig. 4 show the effect of frequency resolution. The 256-bin grid has mismatch 0.0317 against the 4096-bin reference. The 1024-bin grid reduces the mismatch to 8.47×10^{-4} , and the 2048-bin grid reaches 5.89×10^{-5} . We therefore use the 256-bin results only for low-resolution checks, while the 1024-bin results carry the main inference analysis.

TABLE II. Support-aware FMPE experiment tiers. Evaluation counts are per artifact mode; provenance details are retained in the reproducibility records.

Tier	Bins	Events	Samples	ODE steps	Train steps
low-resolution check	256	12	96	16	80
intermediate check	512	64	256	32	600
primary 1024-bin study	1024	256	512	48	2000
2048-bin resolution study	2048	128	256	48	1500

TABLE III. Waveform-resolution diagnostic against a 4096-bin reference over 30–300 Hz using 128 prior draws. Mismatch is phase-maximized but not time-shift-maximized.

Bins	Relative L2	Phase RMS [rad]	Mismatch
64	0.562471	136.647816	0.177342
128	0.395162	123.246203	0.084973
256	0.236611	96.954127	0.031709
512	0.105759	48.762003	0.007561
1024	0.033538	3.370806	0.000847
2048	0.008697	0.000163	0.000059

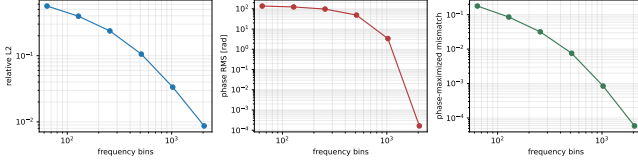


FIG. 4. Waveform-resolution diagnostic against a 4096-bin reference. The 64-, 128-, and 256-bin grids are visibly coarse, while the 1024- and 2048-bin grids are close to the reference under the phase-maximized mismatch used here.

TABLE IV. TaylorF2 backend checks against PyCBC/LAL routines. The controlled-response row uses the same fixed detector gains, delays, and analytic whitening as the simulator.

Check	Metric	Interpretation
Intrinsic phase/amplitude	median phase RMS 0.0272 rad; median mismatch 0.00468	consistent with controlled phase shape
Convention-aligned controlled response	median mismatch 5.23×10^{-4} ; max mismatch 0.0185	consistent with controlled response
Direct PyCBC sky projection	mean relative L_2 error 1.411; mean phase RMS 43.9 rad	out of scope

Table IV summarizes the TaylorF2 backend checks. In the intrinsic comparison, the PyCBC waveform is allowed to differ by arbitrary amplitude scale, constant phase, and time translation. After those convention-dependent factors are removed, the median phase residual is 0.0272 rad and the median mismatch is 0.00468 over 20 prior draws. The controlled detector response then applies the same fixed gains, fixed delays, and analytic whitening as the simulator. In that like-for-like comparison, the median detector-response mismatch is 5.23×10^{-4} and the maximum mismatch is 0.0185. A direct comparison

with a PyCBC sky-response projection uses a different detector-response model and is not used as validation evidence. The simulator is therefore treated as a controlled TaylorF2-like inverse problem throughout the analysis.

B. High-Resolution FMPE Results

TABLE V. Support-aware FMPE runs. MA90CE is mean absolute 90% marginal coverage error with circular phase treatment. Global denotes marginal interval rescaling fitted on a mixed calibration set.

Mode	1024 bins		2048 bins	
	Raw	Global	Raw	Global
clean	0.0305	0.0620	0.0372	0.0644
glitch	0.0458	0.0605	0.0428	0.0625
frequency mask	0.0369	0.1148	0.0559	0.1281
PSD mismatch	0.0436	0.0575	0.0453	0.0581
mixed	0.0319	0.0773	0.0272	0.0734

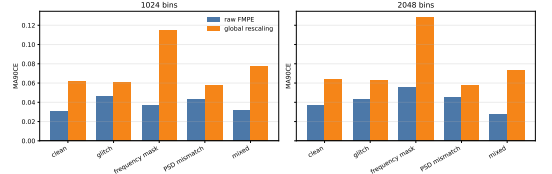


FIG. 5. Raw and globally rescaled MA90CE for the trained 1024-bin and 2048-bin FMPE evaluations. Global marginal rescaling is not reliably beneficial and is especially poor under frequency masking.

The high-resolution FMPE evaluations are summarized in Table V and Fig. 5. At 1024 bins, raw MA90CE ranges from 0.0305 in clean mode to 0.0458 in glitch mode. The 2048-bin extended run is similar, with raw MA90CE from 0.0272 to 0.0559 across modes. In both tiers, global marginal rescaling increases MA90CE for every reported mode. The frequency-mask mode is most affected: global rescaling gives MA90CE 0.1148 at 1024 bins and 0.1281 at 2048 bins.

A scale fitted on mixed calibration data therefore does not transfer uniformly across artifact modes. It can reduce an aggregate coverage error while worsening a particular data-quality condition; in these runs, frequency

masking is the most visible case.

TABLE VI. Standardized 1024-bin FMPE training-seed study across six checkpoints. Values are mean MA90CE with seed-to-seed standard deviation in parentheses. Each checkpoint is evaluated with 256 calibration events, 256 evaluation events per mode, 512 posterior samples per event, and 48 ODE steps. Global denotes marginal interval rescaling fitted on the calibration set.

Mode	Raw FMPE	Global
clean	0.0421 (0.0093)	0.0402 (0.0158)
glitch	0.0448 (0.0092)	0.0362 (0.0167)
frequency mask	0.0443 (0.0128)	0.1175 (0.0271)
PSD mismatch	0.0406 (0.0038)	0.0397 (0.0153)
mixed	0.0334 (0.0082)	0.0393 (0.0222)

Table VI reports a standardized 1024-bin training-seed study across six checkpoints: the main checkpoint, two earlier additional seeds, and three additional checkpoints trained with the same 2000-step protocol. All six are evaluated with 256 calibration events and 256 evaluation events per artifact mode. The study supports the frequency-mask conclusion: raw FMPE has mean frequency-mask MA90CE 0.0443 with seed-to-seed standard deviation 0.0128, whereas global rescaling gives 0.1175 with standard deviation 0.0271. The mixed-mode global result also varies more than the raw result. This study probes training-initialization sensitivity within the controlled benchmark; it is not a substitute for larger production-scale retraining studies.

C. Conditional Normalizing-Flow Baseline

TABLE VII. 1024-bin conditional RealNVP baseline compared with the trained FMPE checkpoint. This is a controlled discrete-flow baseline in the same simulator, not a DINGO reproduction. Entries are MA90CE.

Mode	FMPE		Conditional RealNVP	
	Raw	Global	Raw	Global
clean	0.0305	0.0620	0.0362	0.0541
glitch	0.0458	0.0605	0.0425	0.0806
frequency mask	0.0369	0.1148	0.0728	0.1141
PSD mismatch	0.0436	0.0575	0.0303	0.0419
mixed	0.0319	0.0773	0.0347	0.0619

Table VII adds a 1024-bin conditional RealNVP baseline trained and evaluated in the same simulator. The baseline uses an invertible box-logit phase coordinate for density modeling and decodes phase circularly for metrics. RealNVP is close to FMPE in clean and mixed raw MA90CE, worse under frequency masking, and better in this single run for PSD mismatch. Both model families are sensitive to global rescaling, so the artifact dependence is not unique to the FMPE sampler.

D. Representative Posterior Geometry

Figures 6 and 7 show posterior geometry for the trained 1024-bin checkpoint. They are included because marginal coverage tables do not display correlation structure or multimodality. The likelihood-reference comparison is reported separately in Sec. IIIH; in the checked events it shows substantial FMPE-reference differences.

E. 1024-Bin LAIR Evaluation

Table VIII and Fig. 8 give the 1024-bin artifact-conditioned result. Global rescaling improves clean MA90CE from 0.0452 to 0.0331 and glitch MA90CE from 0.0484 to 0.0169. Under frequency masking, however, the same global scale raises MA90CE from 0.0541 to 0.1195. Soft LAIR gives 0.0273 in glitch mode and reduces the frequency-mask error to 0.0672, but it is worse than the raw FMPE intervals for PSD-mismatch and mixed mode.

The oracle artifact-stratified result is also not uniformly beneficial. Even with true artifact labels, a marginal width correction can move coverage in the wrong direction when the fitted scale is noisy or when the dominant posterior error is not a width error. The method should therefore be read as a mode-resolved marginal coverage diagnostic, not as a correction to posterior sample geometry.

The event-bootstrap intervals in Table IX quantify finite-evaluation uncertainty. For frequency masks, the raw, global, and soft-LAIR intervals are [0.0356, 0.0806], [0.0914, 0.1469], and [0.0470, 0.0930], respectively. For mixed mode, soft LAIR gives MA90CE 0.0594 with interval [0.0416, 0.0767], whereas the raw estimate is 0.0303 with interval [0.0186, 0.0475].

Figure 9 shows the corresponding coverage curves. Rescaling affects nominal levels differently, and the ordering of methods changes across artifact modes.

Table X and Fig. 10 report the repeated calibration study. The 40 repeats used the validation-selected classifier from the classifier comparison, whose balanced accuracy was 0.5029; the study is therefore reported separately from the higher-performing classifier in Table XI. The frequency-mask pattern persists: global rescaling has mean MA90CE 0.1137, while soft LAIR gives 0.0679. In mixed mode, raw FMPE, global rescaling, and soft LAIR have mean MA90CE 0.0323, 0.0327, and 0.0403, respectively.

F. Artifact Classifier

Table XI reports the 1024-bin artifact classifier. Its balanced accuracy is 0.6982, negative log likelihood is 0.5028, and expected calibration error is 0.0489. Frequency-mask recall is 1.0000 and PSD-mismatch recall is 0.9941. Clean recall is 0.5918, and glitch recall

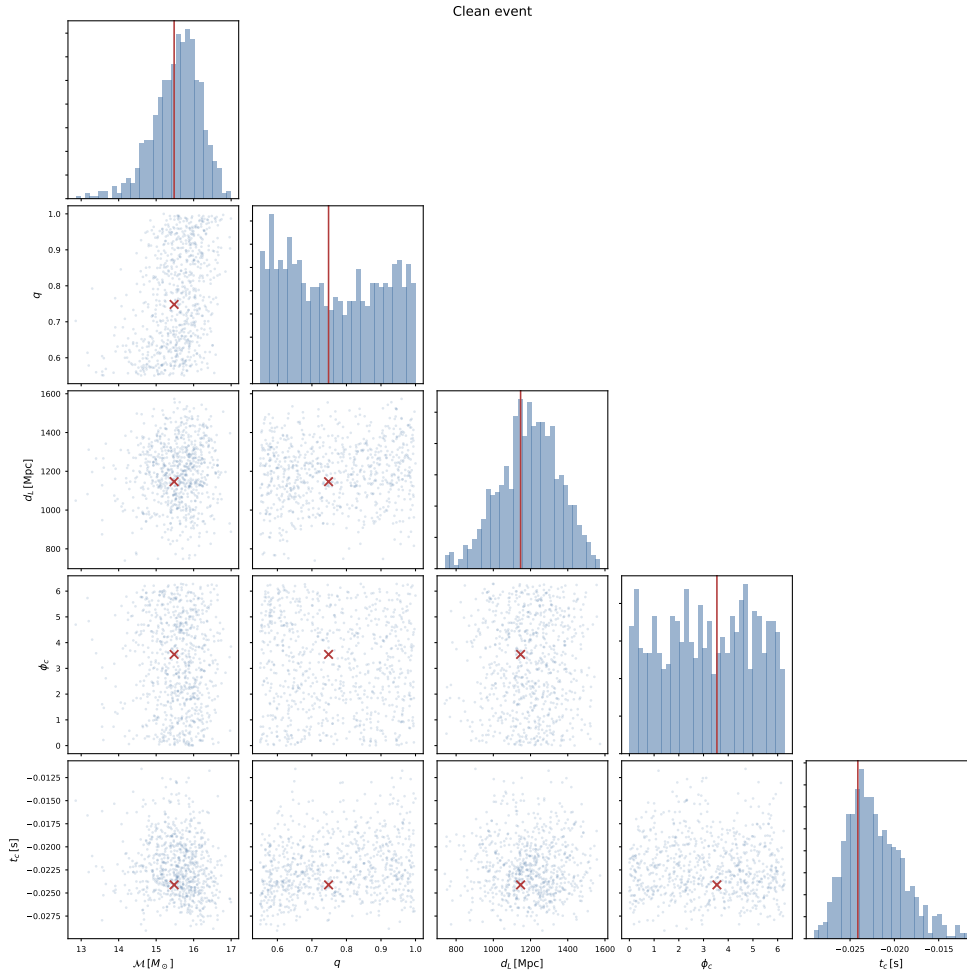


FIG. 6. Representative 1024-bin FMPE posterior geometry for one clean simulated event. The corner plot uses the trained 1024-bin checkpoint and 768 posterior samples. It shows marginal and pairwise structure; no reference posterior is overlaid.

TABLE VIII. 1024-bin LAIR evaluation using the trained 1024-bin FMPE checkpoint, a matching 1024-bin artifact classifier, 256 calibration events, 256 evaluation events per mode, 512 posterior samples per event, 48 ODE steps, and 200 bootstrap resamples. Values are MA90CE.

Mode	Raw FMPE	Global	Oracle	Predicted label	Soft LAIR
clean	0.0452	0.0331	0.0370	0.0447	0.0423
glitch	0.0484	0.0169	0.0705	0.0253	0.0273
frequency mask	0.0541	0.1195	0.0672	0.0672	0.0672
PSD mismatch	0.0397	0.0394	0.0634	0.0486	0.0634
mixed	0.0303	0.0414	0.0453	0.0422	0.0594

is 0.2070. Thus, for this classifier, masks and PSD mismatch are well identified, while glitch recognition is the main limitation for LAIR.

The classifier comparisons test whether residual or focal-loss variants improve this tradeoff.

Model	Bal.	NLL	R_c	R_g	R_m	R_p
baseline	0.6982	0.5028	0.5918	0.2070	1.0000	0.9941
residual+focal	0.5015	0.8247	0.1270	0.6289	1.0000	0.2500
deep+focal	0.5049	0.8241	0.0078	0.0605	1.0000	0.9512
wide residual	0.4980	0.8289	0.8164	0.0508	1.0000	0.1250
validation seed	0.7510	0.3815	0.7715	0.2539	1.0000	0.9785

TABLE XII. Glitch-targeted 1024-bin artifact-classifier comparisons. Bal. is balanced accuracy, NLL is negative log likelihood, and the recall columns correspond to clean, glitch, frequency-mask, and PSD-mismatch classes.

Table XII gives the resulting tradeoff. In the initial three-configuration comparison, a residual width-96 focal-loss run raises glitch recall to 0.6289, but balanced accuracy falls to 0.5015, negative log likelihood increases

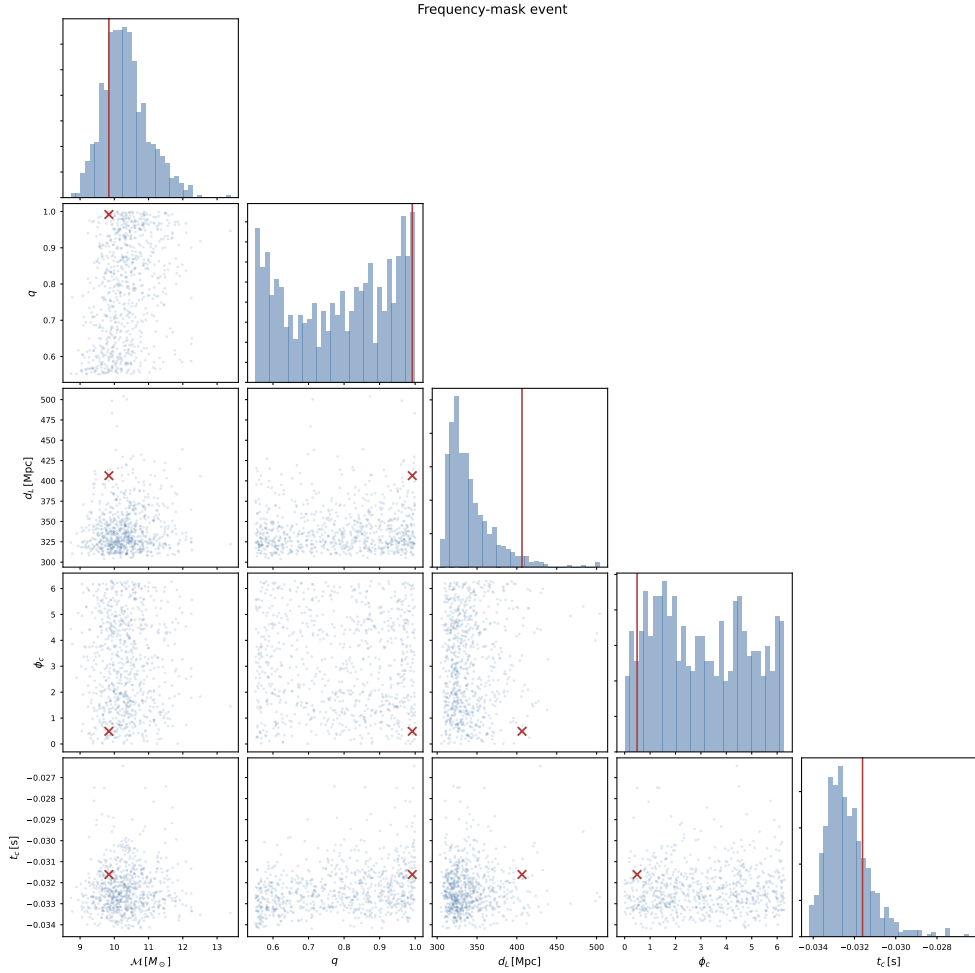


FIG. 7. Representative 1024-bin FMPE posterior geometry for one frequency-mask simulated event, using the same plotting convention as Fig. 6.

TABLE IX. Event-bootstrap 95% intervals for 1024-bin MA90CE. Each entry is estimate [2.5%, 97.5%].

Mode	Raw FMPE	Global	Soft LAIR
clean	0.0452 [0.0334, 0.0567]	0.0331 [0.0190, 0.0502]	0.0423 [0.0267, 0.0610]
glitch	0.0484 [0.0383, 0.0594]	0.0169 [0.0136, 0.0369]	0.0273 [0.0159, 0.0500]
frequency mask	0.0541 [0.0356, 0.0806]	0.1195 [0.0914, 0.1469]	0.0672 [0.0470, 0.0930]
PSD mismatch	0.0397 [0.0289, 0.0530]	0.0394 [0.0268, 0.0560]	0.0634 [0.0439, 0.0814]
mixed	0.0303 [0.0186, 0.0475]	0.0414 [0.0283, 0.0603]	0.0594 [0.0416, 0.0767]

to 0.8247, and clean and PSD-mismatch recall deteriorate. A follow-up residual/focal run with fresh validation seeds reaches balanced accuracy 0.7510 and negative log likelihood 0.3815. It improves clean, frequency-mask, and PSD-mismatch recognition, but glitch recall remains 0.2539.

G. Prior and Gaussian Baselines

TABLE XIII. Diagonal Gaussian posterior baseline on the 256-bin support-aware low-resolution setting. Values are MA90CE.

Mode	Prior only	Gaussian raw	Gaussian global
clean	0.0381	0.0313	0.0650
glitch	0.0706	0.0256	0.1181
frequency mask	0.0194	0.0288	0.0900
PSD mismatch	0.0175	0.0425	0.1094
mixed	0.0263	0.0219	0.0994

Table XIII illustrates the limits of a marginal coverage metric. The prior-only baseline has low MA90CE in several modes because broad central prior intervals often contain the truth. The diagonal Gaussian approximation also has low coverage error in some settings despite its limited posterior structure. Coverage error must there-

TABLE X. Repeated 1024-bin LAIR calibration study across 40 independent simulator/calibration splits using the validation-selected classifier from the classifier comparison. Entries are mean MA90CE with split-to-split standard deviation in parentheses. This classifier is reported separately because it is not the higher-performing classifier in Table XI.

Mode	Raw FMPE	Global	Oracle	Predicted label	Soft LAIR
clean	0.0433 (0.0059)	0.0367 (0.0119)	0.0394 (0.0121)	0.0427 (0.0136)	0.0425 (0.0142)
glitch	0.0432 (0.0055)	0.0353 (0.0085)	0.0593 (0.0225)	0.0401 (0.0108)	0.0420 (0.0138)
frequency mask	0.0521 (0.0096)	0.1137 (0.0165)	0.0679 (0.0202)	0.0679 (0.0202)	0.0679 (0.0201)
PSD mismatch	0.0436 (0.0072)	0.0344 (0.0089)	0.0581 (0.0201)	0.0386 (0.0107)	0.0409 (0.0124)
mixed	0.0323 (0.0068)	0.0327 (0.0100)	0.0380 (0.0111)	0.0374 (0.0107)	0.0403 (0.0123)

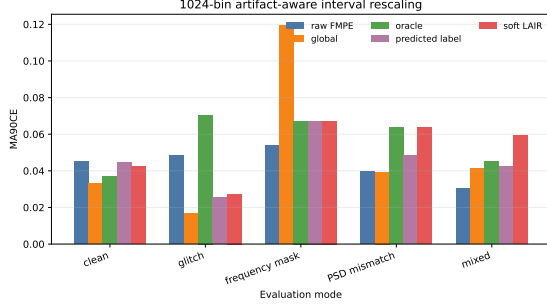


FIG. 8. 1024-bin LAIR comparison using the trained high-resolution FMPE checkpoint and a matching 1024-bin artifact classifier. Rescaling changes the coverage error in a strongly mode-dependent way.

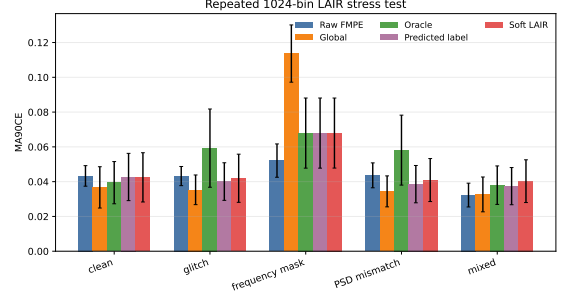


FIG. 10. Repeated 1024-bin LAIR calibration study across 40 independent simulator/calibration splits. Error bars show split-to-split standard deviation.

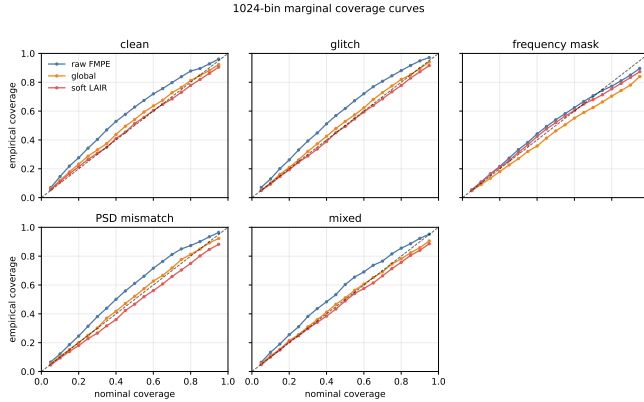


FIG. 9. 1024-bin marginal coverage curves averaged over parameters for raw FMPE, global rescaling, and soft LAIR. The diagonal is ideal calibration. Frequency masking shows the most visible separation between global rescaling and the artifact-aware diagnostic.

fore be considered together with posterior width and geometry.

H. Controlled-Likelihood Reference Checks

A first check compared a quadratic Laplace approximation with FMPE samples for five clean 1024-bin events. The mean sliced Wasserstein distance was 0.3988 and the median Hessian condition number was 2.119×10^5 , indi-

TABLE XI. 1024-bin artifact-classifier validation metrics on 2048 balanced validation events.

Metric	Value
Accuracy	0.6982
Balanced accuracy	0.6982
Negative log likelihood	0.5028
Expected calibration error, 10 bins	0.0489
Clean recall	0.5918
Glitch recall	0.2070
Frequency-mask recall	1.0000
PSD-mismatch recall	0.9941
Clean precision	0.4951
Glitch precision	0.4753
Frequency-mask precision	1.0000
PSD-mismatch precision	0.7261

cating a poorly conditioned local approximation rather than a reliable reference posterior. An exploratory importance-sampling run over three clean events produced median effective sample size 1.0 from 20000 prior proposals. These runs show that the controlled likelihood can be interrogated, but they do not provide validated reference posteriors.

Table XIV reports further controlled-likelihood reference-posterior checks. The dynesty runs cover the target event counts, namely five clean and two frequency-mask 1024-bin observations, but they do not meet the evidence-error gate: the median evidence-error diagnostics are 0.668 and 0.664 for clean and frequency-mask events, respectively. Repeated single-chain Hamiltonian Monte Carlo (HMC) runs also did not pass diagnostics,

TABLE XIV. Controlled-likelihood reference-posterior checks. W_{sliced} and W_{par} are mean sliced and parameter-wise Wasserstein distances in prior-unit coordinates. The dynesty and single-chain HMC rows did not satisfy the stated diagnostics. The final stretch-MCMC rows use a Laplace-preconditioned ensemble proposal and meet the split- $\hat{R}$ and effective-sample-size diagnostics for the target clean and frequency-mask event counts.

Method	Mode	Ev.	W_{sliced}	W_{par}	diagnostic
dynesty	clean	5	0.4399	0.356	median logzerr 0.668
dynesty	frequency mask	2	0.3851	0.347	median logzerr 0.664
HMC	clean	3×5	0.3947–0.4492	–	median acceptance 0.0
HMC	frequency mask	2×2	0.3858–0.4297	–	median acceptance 0.0
stretch MCMC	clean	5	0.3912	0.3301	max $\hat{R}$ = 1.0089, ESS _{min} = 3744
stretch MCMC	frequency mask	2	0.4705	0.3596	max $\hat{R}$ = 1.0091, ESS _{min} = 4182

with median acceptance rate zero in all completed clean and frequency-mask runs.

The most stable reference check uses a Laplace-preconditioned ensemble stretch Markov chain Monte Carlo (MCMC) sampler on the same controlled Gaussian likelihood. For five clean events it gives mean sliced Wasserstein distance 0.3912, mean parameter-wise Wasserstein distance 0.3301, maximum split- $\hat{R}$ = 1.0089, and minimum effective sample size 3744. For two frequency-mask events the corresponding values are 0.4705, 0.3596, 1.0091, and 4182. Postprocessing of the MCMC samples gives mean 90% interval-overlap ratios 0.0097 and 0.0109 for clean and frequency-mask events. The small overlaps indicate substantial differences between the trained FMPE posterior and the controlled-likelihood reference in these examples. This check is a simulator-level reference diagnostic; it is not a Bilby/PyCBC or LVK-production posterior comparison.

IV. DISCUSSION

The 1024-bin experiments show clear artifact dependence in marginal interval calibration. A global scale fitted on mixed calibration data reduces the error in some modes but substantially worsens frequency-mask coverage. Soft LAIR mitigates that failure, lowering the 1024-bin frequency-mask MA90CE from the global value of 0.1195 to 0.0672, but it does not dominate the raw FMPE intervals across all artifact modes. The repeated evaluation gives the same qualitative result across simulator and calibration seeds.

Classifier behavior is part of this outcome. Frequency masks are explicit in the mask channel and are identified with high recall. PSD mismatch is also recognized reliably by the main classifier. Glitches are less well separated from clean observations in the evaluated feature representation. When the classifier assigns little probability to the glitch class, the soft LAIR scale is pulled toward other artifact scales, limiting glitch-conditioned rescaling.

The auxiliary diagnostics constrain the interpretation of coverage. The waveform-resolution test shows that coarse frequency grids can change the inference problem. The prior-only and Gaussian baselines show that low coverage error can arise from broad or structurally sim-

ple intervals. Posterior-geometry plots and controlled-likelihood reference checks further show that marginal coverage does not determine joint posterior structure or agreement with a likelihood-based target.

The support-aware coordinate transform and circular phase representation remove two avoidable implementation artifacts: decoded samples leaving the prior domain and discontinuities at the phase boundary. In this controlled benchmark, the remaining limitations are artifact-dependent interval behavior, weak glitch recognition, and differences between FMPE samples and the controlled-likelihood reference. Extension to production gravitational-wave inference would require stronger transient-artifact classification, multivariate posterior diagnostics, full-budget training seeds, and comparison with Bilby/PyCBC or related likelihood-based analyses in an International Gravitational-Wave Network (IGWN)-compatible environment.

V. CONCLUSIONS

We evaluated flow-matching neural posterior estimation in a controlled gravitational-wave inverse problem with explicit artifact labels. The analysis focused on marginal interval calibration for clean observations, synthetic glitches, frequency masks, and PSD mismatch.

The central empirical observation is mode dependence. In the 1024-bin evaluation, global marginal rescaling improves some modes but raises the frequency-mask MA90CE to 0.1195. Soft LAIR reduces that value to 0.0672, while remaining worse than raw FMPE in PSD-mismatch and mixed mode. The 40-split calibration study supports the same interpretation: frequency masks are a stable failure mode for global rescaling, and artifact-conditioned rescaling is useful primarily as a mode-resolved interval diagnostic.

The classifier results identify a practical bottleneck. Frequency masks and PSD mismatch are recognized reliably by the main 1024-bin classifier, whereas glitch recall is low. The best residual/focal seed improves aggregate validation metrics, reaching balanced accuracy 0.7510, but its glitch recall is still 0.2539. More transient-sensitive features or architectures are needed before glitch-conditioned interval scaling can be relied on.

The supporting checks are necessary for interpreting these numbers. Support-aware coordinates keep bounded parameters inside the prior domain, and the circular phase embedding removes the artificial boundary at 0 and 2π . Waveform-resolution tests motivate the use of 1024- and 2048-bin grids. The prior-only, Gaussian, conditional-flow, Laplace, dynesty, and multi-chain MCMC checks show that marginal coverage alone can be misleading when intervals are broad, joint structure is

wrong, or the neural posterior differs from a likelihood-based reference.

Within this controlled setting, LAIR provides a reproducible way to expose artifact-conditioned interval failures in neural posterior estimation. Moving to production gravitational-wave inference would require stronger glitch classification, multivariate posterior validation, and comparison with production-style likelihood-based analyses on more realistic detector data.

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